

Module 3:

Problem Statement:

We know bKash is a pioneering mobile financial service in Bangladesh that has played a significant role in transforming the country's financial landscape. Launched in 2011, bKash was established as a joint venture between BRAC Bank Limited, one of Bangladesh's leading private banks, and Money in Motion LLC, a US-based technology company. Our goal is to write a similar type of program that would be very close to their functionalities. The name of our application is **MyCash**.

Write a C++ program to create MyCash application using class and objects to fulfill the following criteria.

- i. The program is capable of storing data for each member. The member data includes mobile-no, name, amount and pin.
- ii. To use the system an user must register first.
- iii. The program should be menu operated
- iv. The program starts with a login menu. If login successful then main menu appears.
- v. Define a class with data members mobile, name, amount and pin.
- vi. Treat mobile number as the key information for each member.
- vii. Create facilities for cash-in, cash-out, send money and pay bill.
- viii. The system must check for duplicate member
- ix. The application should produces appropriate error message when required (table 1)
- x. The system must use OTP (one time passcode) when more security is required like Add Member, Change pin, Send money, Cash-out and Pay bill.
- xi. The History displays the details of each transaction.

Code:

```
#include<bits/stdc++.h>
#include<conio.h> // for using getch function
using namespace std;
#define ll long long int
#define str string
#define P_B push_back
#define p_b pop_back
#define pi 3.141592654
int f(int x){if(x==1){return 1;}return x*f(x-1);}
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define A_sort(x) sort(x.begin(),x.end())
#define D_sort(x) sort(x.rbegin(),x.rend())
#define reverse(x) reverse(x.begin(),x.end())
#define endl "\n"
#define newl cout<<endl
#define clr system("cls");
int main();
map<str,int>mp,mp2;
int id(str m)
{
    return mp[m];
}
void menu()
{
    cout<<"*****MyCash menu*****\n";
    cout<<"1. Update Me\n2. Remove Me\n3. Send Money\n4. Cash-in\n5. Cash-out\n6. Pay Bill\n7. Check Balance\n8. History\n9. Logout\n";
    cout<<"Enter Your Option (1-9):_";
    newl;
```

```
}
```

```
*****MyCash menu*****
1. Update Me
2. Remove Me
3. Send Money
4. Cash-in
5. Cash-out
6. Pay Bill
7. Check Balance
8. History
9. Logout
Enter Your Option (1-9):_
```

```
bool m_check(str s)
{
    int n=s.size();
    for(int i=0;i<n;i++)
    {
        if(s[i]>='0'&&s[i]<='9'&&s[0]=='0'&&s[1]=='1')
        {
            continue;
        }
        else
        {
            return false;
        }
    }
    return true;
}

void login()
{
    cout<<"*****MyCash Login*****\n";
    cout<<"1. Login\n2. Register\n3. Exit\nEnter Your Option: ";
}
```

```
*****MyCash Login*****
1. Login
2. Register
3. Exit
Enter Your Option: |
```

```
int OTP_gen()
{
    int x; srand(time(0));
    x=1000+rand()%9000; return x;
}

str pass_gen()
{
    str p="";
    char ch; ch=getch();
    while(ch!=13)
    {
        p.P_B(ch);
        cout<<"*";
        ch=getch();
    }
    newl; return p;
}

int total=1;
int member=0;
```

```

class myCash
{
    private:
        str name,mobile,pin;
        double balance;
        vector<pair<str,int>>his;
    public:
        myCash()
        {

        }
        void set_info(str m,str p,str n,double b)
        {
            mobile=m; pin=p; name=n; balance=b;
        }
        void login(str n, str p)
        {
            if(n==mobile&&p==pin)
            {
                if(mp2[mobile]>0)
                {
                    balance+=mp2[mobile]; mp2[mobile]=0;
                }
                jao5:
                clr  menu();
                int a; cin>>a;
                if(a==1){update();}
                else if(a==2){remove();}
                else if(a==3){send();}
                else if(a==4){cash_in();}
                else if(a==5){cash_out();}
                else if(a==6){pay_bill();}
                else if(a==7){c_balance();}
                else if(a==8){History();}
                else if(a==9){logout();}
                else
                {
                    cout<<"Invalid option\n";
                    cout<<"\n Press any key to continue\n";
                    getchar();getchar();
                    clr  goto jao5;
                }
            }
            else
            {
                cout<<"password or mobile number is incorrect\nPress any key to continue ";
                getchar();getchar();
                clr  main();
            }
        }
    private:
        void update()
        {
            jao2:
            cout<<"Enter your option\n";
            cout<<"1. Update name\n2. Update Pin\n";
            cout<<"Your option-> ";
            int a; cin>>a; getchar();
            if(a==1)

```

```

{
    cout<<"Old name->"<<name;newl;
    cout<<"New name->";
    getline(cin,name);
    cout<<"Name Upadted Succesfully\n";
    cout<<"Press enter to continue";
    getchar();
    login(mobile,pin);
}
else if(a==2)
{
    cout<<"Old pin->"<<pin;newl;
    cout<<"New Pin->";
    str s=pass_gen();
    if(s.length()==5)
    {
        pin=s;
        cout<<"pin updated succcesfully\n";
        cout<<"Press enter to continue";
        getchar();
        clr login(mobile,pin);
    }
    else
    {
        cout<<"size must be 5";
        cout<<"press enter to continue\n";
        getchar(); getchar();
        clr goto jao2;
    }
}
else
{
    cout<<"you have entered incorrect key\n";
    cout<<"press enter to continue";
    getchar(); getchar();
    clr goto jao2;
}
}
}

```

```

Enter Your Option (1-9):_
1
Enter your option
1. Update name
2. Update Pin
Your option-> 1
Old name->Adil.
New name->Adil
Name Upadted Succesfully
Press enter to continue

```

```

void remove()
{
    jao6:
    clr
    cout<<"\nARE YOU SURELY WANT TO DELETE YOUR ACCOUNT\n";
    cout<<"Press Y for YES or N for NO: ";
    char c; cin>>c;
    if(c=='Y' | | c=='y')
    {
        mp.erase(mobile);
        cout<<"Successfully Removed Account\n";
    }
}

```

```

        cout<<"Press any key to continue\n";
        member--; getchar(); getchar();
        clr main();
    }
    else if(c=='N' || c=='n')
    {
        clr login(mobile,pin);
    }
    else
    {
        cout<<"Invalid Keyword\nPress any key to continue ";
        getchar(); getchar(); goto jao6;
    }
}

```

```

Enter Your Option (1-9):_
2|

ARE YOU SURELY WANT TO DELETE YOUR ACCOUNT
Press Y for YES or N for NO: Y
Successfully Removed Account
Press any key to continue
|

```

```

void send()
{
    cout<<"Enter mobile number: ";
    str n; cin>>n;
    if(mp[n])
    {
        cout<<"Enter Amount: ";
        int l;
        cin>>l;
        cout<<"Enter your pin: ";
        str r=pass_gen();
        if(pin==r)
        {
            mp2[n]+=l; balance-=l;
            cout<<"Send money to "<<n<<" is successful\nPress enter to continue\n";
            his.P_B({n+" has received money of ",l});
            getchar();getchar();
            clr login(mobile,pin);
        }
        else
        {
            cout<<"pin is incorrect\nPress enter to continue\n";
            getchar();getchar();
            clr login(mobile,pin);
        }
    }
    else
    {
        cout<<"Mobile number not found\n";
        cout<<"Press any key to continue ";
        getchar();
        getchar();
        clr
        login(mobile,pin);
    }
}
}

```

```

Enter Your Option (1-9):_
3
Enter mobile number: 01632039154
Enter Amount: 2500
Enter your pin: *****
Send money to 01632039154 is successful
Press enter to continue

```

```

void cash_in()
{
    cout<<"Enter amount ";
    int l;
    cin>>l;
    this->balance+=l;
    his.P_B({"You have cash in the money ",l});
    cout<<"Cash-in Succesfull\n";
    cout<<"Press any key to continue ";
    getchar(); getchar();
    clr login(mobile,pin);
}

```

```

Enter Your Option (1-9):_
4
Enter amount 4000
Cash-in Succesfull
Press any key to continue |

```

```

void cash_out()
{
    cout<<"Enter amount: ";
    int l; cin>>l;
    if(l>balance)
    {
        cout<<"Not enough amount\n";
        cout<<"Press any key to continue: ";
        getchar(); getchar();
        clr login(mobile,pin);
    }
    this->balance-=l;
    his.P_B({"You have cash out the money ",l});
    cout<<"Cash-out Succesfull\n";
    cout<<"Press any key to continue ";
    getchar(); getchar();
    clr login(mobile,pin);
}

```

```

Enter Your Option (1-9):_
5
Enter amount: 1000
Cash-out Succesfull
Press any key to continue |

```

```

void pay_bill()
{
    cout<<"Enter pay bill id: ";
    str r; cin>>r;
    cout<<" Enter amount: ";
    int l; cin>>l;
    if(l>balance)
    {

```

```

        cout<<"Not enough amount\n";
        cout<<"Press any key to continue: ";
        getchar(); getchar();
        clr login(mobile,pin);
    }
    this->balance-=l;
    his.P_B({"You have pay bill the money ",l});
    cout<<"Bill Pay to id "<<r<<" is Succesfull\n";
    cout<<"Press any key to continue ";
    getchar(); getchar();
    clr login(mobile,pin);
}

```

```

Enter Your Option (1-9):_
6
Enter pay bill id: 1234
Enter amount: 500
Bill Pay to id 1234 is Succesfull
Press any key to continue |

```

```

void c_balance()
{
    cout<<"Your Account Balance is "<<balance;
    newl;
    cout<<"Press any key to continue ";
    getchar(); getchar();
    clr login(mobile,pin);
}

```

```

Enter Your Option (1-9):_
7
Your Account Balance is 14700
Press any key to continue |

```

```

void History()
{
    if(his.size()==0)
    {
        cout<<"There is no History to show right now";
        newl;
        cout<<"Press any key to continue ";
        getchar(); getchar();
        clr login(mobile,pin);
    }
    int i=1;
    for(auto it:his)
    {
        cout<<i<<" ";
        cout<<it.first<<it.second;
        newl;
        i++;
    }
    cout<<"Press any key to continue ";
    getchar();
    getchar();
    clr login(mobile,pin);
}

```

```

Enter Your Option (1-9):_
8
1. 01632039154 has received money of 2500
2. You have cash in the money 4000
3. You have cash out the money 1000
4. You have pay bill the money 800
Press any key to continue |

```

```

void logout()
{
    cout<<"You have succesfully logged out\nPress any key to continue ";
    getchar(); getchar();
    clr main();
}

```

```

Enter Your Option (1-9):_
9
You have succesfully logged out
Press any key to continue |

```

```

};
vector<myCash>members(1000);
int main()
{
    start:
    login();
    while(1)
    {
        int x; cin>>x;
        if(x==1)
        {
            clr
            cout<<"Enter your phone number: ";
            str n; cin>>n;
            if(mp[n])
            {
                cout<<"Enter your pin: ";
                str p=pass_gen();
                members[mp[n]].login(n,p);
            }
            else
            {
                cout<<"Phone Number not registered\n";
                cout<<"Press any key to continue\n";
                getchar(); getchar();
                clr goto start;
            }
        }
    }
    else if(x==2)
    {
        jao:
        clr str mobile;
        cout<<"Enter your mobile number: \n";
        cin>>mobile;
        if(mobile.size()!=11&&m_check(mobile))
        {
            cout<<"Mobile number is incorrect\npress enter to continue";
            getchar(); getchar();
            goto jao;
        }
    }
}

```



```

else if(mp[mobile])
{
    cout<<"mobile number is already registered\npress enter to continue";
    getchar(); getchar();
    clr main();
}
else
{
    str name;
    cout<<"Enter your name: ";
    cin>>name;
    cout<<"Enter pin (5-digits): ";
    str r=pass_gen();
    if(r.size()!=5)
    {
        cout<<"Pin must be 5 digit\nPress enter to continue";
        getchar(); getchar();
        clr main();
    }
    cout<<"Reconfirm pin: ";
    str pin=pass_gen();
    if(pin==r&&pin.size()==5)
    {
        int a=OTP_gen();
        cout<<"YOUR MYCASH OTP---> "<<a;
        newl;
        cout<<"Enter your OTP which is send to "<<mobile;newl;
        int b;cin>>b;
        if(b==a)
        {
            cout<<"Enter amount of first deposit: ";
            double balance;
            cin>>balance;
            cout<<"Registration succesfull\n";
            cout<<"Press any key to continue\n";
            mp[mobile]=total;
            members[mp[mobile]].set_info(mobile,pin,name,balance);
            total++; member++;
            mp2[mobile]=0; getchar(); getchar();
            clr main();
        }
        else
        {
            cout<<"registartion failed\n";
            cout<<"press enter to continue ";
            getchar(); getchar();
            clr main();
        }
    }
}
}

```

```

Enter your mobile number:
01779938352
Enter your name: Asad
Enter pin (5-digits): *****
Reconfirm pin: *****
YOUR MYCASH OTP---> 8090
Enter your OTP which is send to 01779938352
8090
Enter amount of first deposit: 15000
Registration succesfull
Press any key to continue

```

```
}  
else if(x==3)  
{  
    return 0;  
}  
else  
{  
    cout<<"Invalid Key\nPress any key to continue";  
    getchar();  
    getchar();  
    main();  
}  
}  
}
```